Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/07/2008
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Project Title:	The Impact of HZE Particles on Adult Neural Stem Cells and Neurogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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Flight Program:			
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COI Name (Institution):	Eisch , Amelia (U TX Southwestern)		
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Task Description:	Astronauts in deep-space missions are exposed galactic cosmic radiation, which consists of HZE particles. Both the1996 National Academy of Science report of interplanetary mission crew hazards and the NASA Radiation Health Bioastronautics Roadmap emphasize that estimating radiation risks to the central nervous system is a high research priority. One CNS cell population that warrants careful analysis for their response to radiation is adult neural stem cells, since these pluripotent cells may play an important role in recovery after radiation-induced brain injury. Studies from radiation biology and tumor biology have generated mixed information whether adult neural stem cells are radiosensitive or not. Understanding how adult neural stem cells respond to radiation and identification of the characteristics of the radioresistant neural stem cells will be pivotal in crafting future strategies to harness stem cells for repair of the injured brain. In this proposed study, We will perform a comprehensive analysis of the response and potential recovery of adult		

Task Description:	neural stem cells after HZE radiation using two mouse models: 1) the nestinGFP transgenic mouse, which allows visualization of the neural stem cell population within the adult brain at any point in time; and 2) our unique nestin-CreERT2/R26R-YFP transgenic mouse, which allows inducible and permanent YFP labeling of adult neural stem cells and their progeny for the remainder of the cell's life. These complementary models allow us to track the proliferation, differentiation, and survival of neural stem cells and their progeny in vivo after HZE radiation. We hypothesize that adult neural stem cells maintain some ability to repopulate the injured brain after radiation, and that this ability improves with time. We further hypothesize that these invulnerable stem cells will provide clues to radioresistance, which may involve the multiple strategies that neural stem cells employ to repair damaged DNA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research targets several questions on NASA's Radiation Health Bioastronautics Roadmap, all in the category of Acute and Late CNS Risks. The proposed research utilizes two novel transgenic mouse models for a comprehensive analysis of adult neural stem cells and their progeny in vivo for the temporal events of cells survival, proliferation, and differentiation in response various low- or high-LET radiation. The resulting information on the long-term neurogenic and/or degenerative activities will provide accurate assessments of CNS risks after space radiation exposure. The results from Aim 3 in particular will provide strong indication whether adult neural stem cells can even partially recover from radiation injury and produce mature neurons. This approach can also identify the threshold of radiation injury to adult neurogenic potential. Understanding how adult neural stem cells respond to radiation will allow assessment of the tantalizing possibility that the remaining neural stem cells to repopulate the more rapidly dividing progenitor cells and immature neurons that are sensitive to radiation. Even if the remaining neural stem cells are not able to repopulate the cell types lost after irradiation, identification of the characteristics of the radioresistant neural stem cells will itself be pivotal in crafting future strategies to harness stem cells for repair of the injured brain.
Task Progress:	During this first year, we were able to make good progress on our proposed X-ray radiation studies, and those results are summarized here. However, we also were able to send our first group of mice to Brookhaven National Labs for exposure to cosmic radiation, and look forward to analyzing that data. Summary of X-ray work thus far: 1. Replicating work by others, x-ray irradiation causes a long-lasting decrease in hippocampal proliferation. 2. There are no sex differences in basal or radiation-induced deficits in hippocampal proliferation. 3. Voluntary exercise can ameliorate the radiation-induce deficits in proliferation, regardless of delay after irradiation. 4. Neither voluntary exercise alone, irradiation alone, or voluntary exercise after irradiation change the number of Type 1 stem like cells. This supports out hypothesis of the stability of neural stem cells in vivo, which is in sharp contrast to the regulation of proliferating cells by these stimuli. 5. However, age at the time of irradiation may influence Type 1 radiosensitivity. Overall, our current data clearly indicate the differential regulation of proliferating versus stem cells in the adult mouse brain. Future work will expand on these findings to include out data from the mice irradiated with cosmic radiation; a time course to analyze the age-dependence of the radiosensitivity of Type 1 stem cells; and additional markers to evaluate the impact of these stimuli on actual neurogenesis.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Abstracts for Journals and Proceedings	Eisch AJ, DeCarolis NA, Arguello AA, Chen BP. "Adult hippocampal neural stem cells in vivo: impact of radiation and voluntary exercise." Oral presentation given at the NASA Radiation Investigators Workshop, Philadelphia, PA, June 30-July 3, 2008. NASA Radiation Investigators' Workshop, Philadelphia, PA, June 30-July 3, 2008. , Jul-2008
Abstracts for Journals and Proceedings	DeCarolis NA, Arguello AA, Han H, Ables JL, Chen BP, Eisch AJ. "Stem cells in the adult male and female brains: sensitivity to radiation and voluntary exercise." Poster presentation given at the NASA Radiation Investigators Workshop, Philadelphia, PA, June 30-July 3, 2008. NASA Radiation Investigators' Workshop, Philadelphia, PA, June 30-July 3, 2008. , Jul-2008
Abstracts for Journals and Proceedings	Eisch AJ, DeCarolis NA, Arguello AA, Chen BP. "Adult hippocampal neural stem cells in vivo: impact of radiation and voluntary exercise." Oral presentation given at the Cancer Center Mini-Retreat on Radiation Biology, UT Southwestern Medical Center, Dallas Texas, March 4, 2008. Cancer Center Mini-Retreat on Radiation Biology, UT Southwestern Medical Center, Dallas Texas, March 4, 2008. , Mar-2008